

**UKFN Special Interest Group in Nuclear Thermal Hydraulics (SIG-NTH) and
Collaborative Computational Project for Nuclear Thermal Hydraulics (CCP-NTH)**

Annual Technical Meeting

29th and 30th June 2026

Whitworth Art Gallery, Oxford Rd, Manchester M15 6ER

Monday 29th June

10:30 - 11:00 Registration & Refreshments

11:00 - 11:15 Welcome Remarks – *Hector Iacovides*

11:15 - 12:00 The UK NTH Community Today

Round the room ~ 45sec introduction from each attendee

12:00 - 12:30 **Invited Talk – TBC – Sanjeev Gupta, Framatome**

12:30 - 13:30 Lunch

Session 1 Community and Strategy

13:30 - 13:50 CCP-NTH Overview - Shuisheng He, The University of Sheffield

13:50 - 14:10 CCP-NTH Research software development and maintenances - Wei Wang, STFC

Session 2 National Nuclear Strategy, Funding and Perspectives

14:10 - 14:30 **Title(TBC) - Daniel Mathers, Executive Director, Nuclear Innovation and Research Office (NIRO)**

14:30 - 14:50 **Title(TBD) - Afia Masood, Portfolio Manager, EPSRC, UKRI**

14:50 - 14:55 Current Research Topics (*flash talk*) - Sam Mason, Government

Session 3 Experimental Facilities and Safety

14:55 - 15:15 Experimental investigation of passive natural circulation cooling for Generation IV nuclear reactors - Mohammad Ferdoosi, The University of Manchester

15:15 - 15:35 Updates from the BULLET and THOR facilities at Bangor University - Walter Villanueva, Bangor University

15:35 - 15:55 Progress in higher-fidelity experimental measurements of flow boiling at RR - Brodie Upton, Rolls-Royce Submarines

15:55 - 16:00 Hydrogen Safety in Nuclear Powerplant (*flash talk*) - Yang Zhang, The University of Sheffield

16:00 - 16:30 Coffee Break & Group Photo (for those who gave photo permission)

Session 4 Thermal-Hydraulics Modelling and Industrial Applications

16:30 - 16:50 Development of numerical modelling approaches for accurate prediction of Natural Circulation Loops - Reuben Petty, The University of Manchester

16:50 - 17:10 Developing a two-phase flow coarse grid model for nuclear reactor thermal-hydraulics – Jeggathishwaran, The University of Sheffield

17:10 - 17:30 System TH Modelling – Challenges and Opportunities in Industry - Christopher Ince, Rolls-Royce PLC

17:30 - 17:35 Transient Analysis of a High-Temperature Gas-Cooled Micro-Reactor for Fission Surface Power Applications using Simulink and NPSS (*flash talk*) - David Barton, Rolls-Royce PLC

17:35 - 17:40 Validation of CFD Modelling for Natural Circulation Loops (*flash talk*) - Thomas John, Rolls-Royce PLC

19:00

Dinner

Manchester Museum, Coupland Street (under the arch on Oxford Road and to the left)



Tuesday 30th June

08:30 - 09:00

Arrival tea & coffee

Session 5 Data-Driven, Reduced-Order and Multiscale Modelling

- 09:00 - 09:20 Data-driven boiling predictions: introducing the neural-gated mixture of engineering models approach - Omid Bidar, The University of Sheffield
- 09:20 - 09:40 Reduced-Order Model with Geometry-Conforming, Voxel-Dominated Meshes for Neutron Transport - Andrew Buchan, Queen Mary University of London
- 09:40 - 10:00 TBD - Jakov Popov, Queen Mary University of London
- 10:00 - 10:20 Application of Multiscale Multifidelity Methods for Nucleate Boiling in the Nuclear Industry - Matthew Val Meyers, The University of Manchester
- 10:20 - 10:25 Planned developments in coupled system code and CFD through source terms (*flash*) — Patricia Silva, University of Manchester

10:25 - 10:50

Coffee Break

Session 6 Fundamental Physics and HPC

- 10:50 - 11:20 **Invited talk:** Simulating Tokamak Plasmas for Magnetic Fusion: A CFD-Based Modeling Approach - Eric Serre, CNRS M2P2
- 11:20 - 11:40 The production of plasma turbulence via magnetic reconnection - Nick Williams, The University of Manchester
- 11:40 - 12:00 GPU porting of the DNS solver CHAPSim2 – Bo Liu, STFC UKRI

12.00 – 13:00 *Lunch*

Session 7 Gen-IV Reactor Thermal Hydraulics

- 13:00 - 13:20 **Invited Talk** – The NTH Nuclear Innovation Programme outcomes and Gen-IV UK landscape – Richard Underhill, Frazer-Nash Consultancy
- 13:20 - 14:20 **Panel Discussion: Gen-IV Reactor Thermal Hydraulics and Associated Challenges**
Facilitator: Mike Bluck (ICL)
Panellists: Chris Sigournay (EDF), Richard Underhill (FNC), Shuisheng He (University of Sheffield), Steve Graham (UKNNL), Mike Bluck (ICL), Sanjeev Gupta (TBC, Framatome)

14:20 – 14:30 **Closing Remarks – Shuisheng He**

Conference Notes

- **Enquiries:** Wei Wang (wei.wang@stfc.ac.uk); Alex Skillen (alex.skillen@manchester.ac.uk)
- **Speakers:** Most speaking slots are 20 minutes. Please aim to present for **15 minutes**, allowing **5 minutes** for audience questions and a smooth transition to the next speaker.
- **Accommodation:** Accommodation for academic attendees and CCP-NTH Advisory Committee members requiring a stay on the night of 29 June 2026 has been arranged and pre-paid at: Premier Inn Manchester CC Princess Street. 2 Brook St, Manchester M1 7BJ. Please check in under your own name. Breakfast is included.
- **Conference Dinner**
 - Venue: Manchester Museum, Coupland Street (under the arch on Oxford Road and to the left)
 - Alcohol Policy Notice: In accordance with UKRI funding guidelines, STFC/EPSCRC funding associated with CCP-NTH cannot be used to purchase alcoholic beverages. Alcohol will therefore not be provided as part of the conference dinner.



